

# Amal Omrani

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in [amal-omrani](#)



## Profile

PhD candidate in financial mathematics working on numerically implementable pricing and hedging methods under transaction costs. Experienced in machine learning and Gaussian processes, with a focus on reducing computational and memory complexity.

## Research

### PhD Thesis

2023–Present

**Université Paris Dauphine, CEREMADE**

Dynamic programming and numerical computation of superhedging prices under proportional transaction costs. Supervisor: Prof. Emmanuel Lepinette.

### Research Master

2023

**Master Projects – LIMOS, Université Clermont Auvergne**

Clustering approaches based on reduced Gaussian processes for regression and classification (focus on reducing computational and memory complexity).

GitHub

Supervised by: Prof. Chafik Samir    Co-supervised by: Anis Fradi

### Publications

2025

Lepinette, E., & Omrani, A. Explicit Recursive Construction of Super-Replication Prices under Proportional Transaction Costs, *arXiv:2503.02419*

2024

Omrani, A., Fradi, A., & Samir, C. Reduced Run-time and Memory Complexity Regression with a Gaussian Process Prior, In *JDS*

2023

Ben Amor, M. A., & Omrani, A. Chernoff's inequality in Riesz spaces, *Quaestiones Mathematicae*

### Seminars and Conferences

SMFP-2025, SCF-2025, WQF-2025, Bachelier-2025, JDS-2024, CIMPA-2024, RGOSA-2023

## Teaching

2025–2026

Introduction to Finance, Université Paris Dauphine – Financial markets and derivatives, European options (30 students, 40h)

2024–2025

Statistics L2, Université Paris Dauphine – Probability, sampling and estimation (30 students, 50h)

2024–2025

AI Projects, Université Paris Dauphine – Data analysis (Python) (30 students, 15h)

2019–2023

Algebra I and II, Preparatory Institute for Engineering Studies, Monastir (60 students/year, 18h/week)

## Education

2023–Present

PhD in Financial Mathematics, Université Paris Dauphine, CEREMADE

*Dynamic programming and implementation of financial models from economics and finance*

2022–2023

Master in Mathematics and Applications, Université de Monastir, Tunisia

Thesis and internship at LIMOS Laboratory, Université Clermont Auvergne

2019

Mathematics Agrégation — Probability and Statistics, ENS Tunis

## Skills

Technical

Python, MATLAB,  $\LaTeX$

Soft skills

Scientific communication, mentoring, teamwork, organization

Organization

Co-organizer of Young Researchers' Days at CEREMADE (YRSD 2024–2025)

Languages

French (fluent), English (fluent), Arabic (fluent), Italian (basic)